



Machine Learning

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An Explainable Machine Learning Framework for Customer Churn Prediction Using Imbalanced Classification Techniques

Abstract

Customer churn prediction has become an important application of machine learning in customer relationship management, enabling organisations to identify customers likely to discontinue a service and implement targeted retention strategies. However, churn datasets are often highly imbalanced, making minority class prediction difficult and reducing the effectiveness of traditional classification models. This study develops and evaluates an explainable machine learning framework for customer churn prediction using multiple classification algorithms, including Logistic Regression, K-Nearest Neighbour (KNN), Naive Bayes, and Extreme Gradient Boosting (XGBoost). Data preprocessing techniques including encoding, feature scaling, and Synthetic Minority Oversampling Technique (SMOTE) were applied to improve model performance and address class imbalance. Feature selection and correlation analysis were conducted to identify the most relevant predictors contributing to churn behaviour. Model evaluation was performed using accuracy, precision, recall, F1-score, confusion matrices, ROC analysis, and cross-validation. Hyperparameter tuning using GridSearchCV was further implemented through multiple iterative experiments to improve predictive performance. In addition, Local Interpretable Model-Agnostic Explanations (LIME) was applied to enhance model interpretability and explain individual predictions. The findings showed that XGBoost achieved strong overall performance, while recall for the minority churn class remained the primary optimisation objective due to the business importance of churn detection. The study demonstrates that explainable and carefully tuned machine learning models can provide effective support for customer churn prediction despite the inherent challenges associated with imbalanced datasets.

1. Introduction

Customer churn prediction is an important application of machine learning in customer analytics and decision support systems. Churn occurs when customers discontinue a service, leading to financial loss and reduced retention. In competitive industries such as telecommunications and banking, retaining customers is often more cost-effective than acquiring new ones. Consequently, organisations increasingly use predictive analytics to identify customers likely to churn and support proactive retention strategies.

Machine learning algorithms such as Logistic Regression, K-Nearest Neighbour (KNN), Naive Bayes, Support Vector Machines, and ensemble methods are widely applied in churn prediction because of their ability to identify patterns in large datasets (Géron, 2022; Witten et al., 2016). However, churn datasets are often imbalanced, where non-churn cases outnumber churn cases, causing models to favour the majority class and reduce minority class detection (He & Garcia, 2013). Studies have shown that accuracy alone is insufficient for evaluating imbalanced classification problems, making metrics such as precision, recall, and F1-score more appropriate (Kuhn & Johnson, 2018).

This study develops an explainable machine learning framework using preprocessing, SMOTE balancing, hyperparameter tuning, cross-validation, and LIME explainability to improve churn prediction performance and interpretability (Molnar, 2022).

2. Data Exploration and Feature Engineering

2.1 Overview

This section presents the exploratory data analysis and feature engineering processes applied to the Telco Customer Churn dataset. The objective was to understand the dataset structure, identify data quality issues, and prepare the data for effective machine learning modelling.

2.2 Dataset Inspection

The dataset consists of 7,043 customer records and 21 features, with each row representing an individual customer and each column describing specific attributes related to demographics, services, and account information. The target variable, Churn, indicates whether a customer has discontinued the service.

Initial inspection revealed a combination of categorical and numerical variables. The target variable was binary, represented as Yes or No, making it suitable for classification tasks.

2.3 Target Variable Distribution

Analysis of the target variable revealed significant class imbalance, where non-churn customers outnumbered churn cases, potentially biasing models toward majority class prediction and reducing churn detection.

2.4 Data Preprocessing

Data preprocessing included encoding categorical variables and standardising numerical features to improve compatibility with machine learning algorithms and prevent scale dominance (Géron, 2022). The dataset was split into training and testing sets. Due to class imbalance favouring non-churn cases, SMOTE was applied to improve minority churn representation (He & Garcia, 2013).

2.5 Correlation Analysis

Correlation analysis examined relationships between predictors and identified potential feature redundancy. Moderate correlations were observed among some service-related variables, particularly tenure and total charges, indicating overlapping behavioural patterns. However, no severe multicollinearity was detected, suggesting the feature set remained suitable for classification modelling.

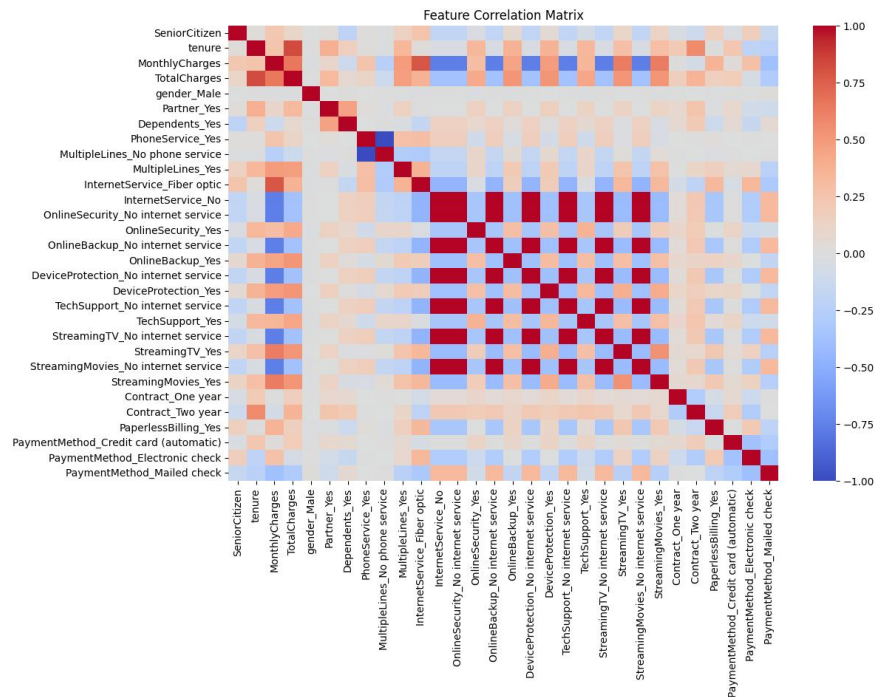


Fig. 1 correlation matrix

The correlation matrix identified relationships between features linked to churn behaviour. Tenure showed a strong positive correlation with TotalCharges, while MonthlyCharges moderately correlated with internet and streaming services. Strong relationships also existed among internet-dependent features due to shared service requirements. Negative correlations appeared between InternetService_No and related services. Overall, mostly weak to moderate correlations were observed, with no severe multicollinearity, suggesting churn is influenced by combined service, billing, and contract factors

2.6 Encoding of Categorical Variables

Categorical variables were transformed into numerical representations using dummy encoding, enabling machine learning models to interpret categorical relationships effectively despite increased dimensionality.

2.7 Feature Selection

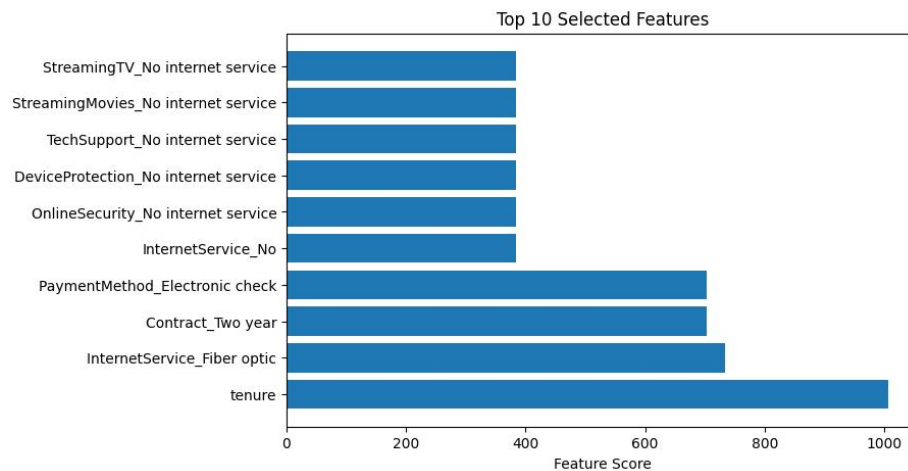


Fig 2. Feature Selection

Feature selection using SelectKBest identified tenure as the strongest predictor of churn, indicating longer-serving customers were less likely to leave. Other important features included InternetService_Fiber optic, Contract_Two year, and PaymentMethod_Electronic check. Internet-related variables also showed strong predictive importance. Overall, churn was mainly influenced by service usage, customer engagement duration, and contract-related factors rather than demographic characteristics.

2.8 Train-Test Split

To evaluate model performance effectively, the dataset was divided into training and testing subsets using an 80:20 ratio. The training set was used to build the models, while the testing set was reserved for evaluating performance on unseen data.

This approach helps ensure that the results are generalisable and not overly fitted to the training data.

2.9 Feature Scaling

Feature scaling standardised numerical variables after data splitting to prevent large-value features dominating KNN and SVM learning. Following dummy encoding, the dataset expanded to 30 features and was divided into 5,625 training and 1,407 testing samples, improving model stability and classification reliability.

3. Methodology

This study applies Logistic Regression, K-Nearest Neighbours (KNN), Naïve Bayes, and Extreme Gradient Boosting (XGBoost) for customer churn prediction. These models were selected for their interpretability, theoretical strengths, and effectiveness in classification tasks within similar domains.

3.1 Logistic Regression

Logistic Regression models the probability of binary outcomes using a logistic function, making it suitable and interpretable for churn prediction (James et al., 2021). It is widely used as a baseline model because of its simplicity, efficiency, and ability to explain feature influence on customer behaviour (Dritsas & Trigka, 2022).

3.2 K-Nearest Neighbours (KNN)

KNN is a non-parametric algorithm that classifies data using the majority class of nearest neighbours, enabling detection of local patterns and non-linear relationships (Acito, 2023; Theobald, 2024). It is widely used for its simplicity and effectiveness in churn prediction despite its computational cost (Acito, 2023).

3.3 Support Vector Machine (SVM)

Support Vector Machine (SVM) identifies an optimal hyperplane for class separation and models linear and non-linear relationships using kernel functions (Géron, 2022; Bishop & Bishop, 2024). It is effective for high-dimensional churn prediction problems but is sensitive to parameter tuning, feature scaling, and computational cost (Géron, 2022; Bishop & Bishop, 2024).

3.4 Naive Bayes

Naïve Bayes is a probabilistic classifier based on Bayes' theorem that performs efficiently in high-dimensional classification tasks and provides a reliable churn prediction baseline (Rish, 2001).

3.4 Extreme Gradient Boosting (XGBoost)

XGBoost is an advanced gradient boosting algorithm that builds sequential decision trees to improve predictive accuracy (Géron, 2022; Raschka & Mirjalili, 2022). Its regularisation capabilities reduce overfitting, making it highly effective for structured customer churn classification tasks.

3.5 Cross-Validation

Cross-validation evaluated model stability and generalisation across multiple folds using accuracy, recall, and F1-score. Logistic Regression showed stable accuracy, while KNN achieved stronger minority-class recall performance.

3.6 Hyperparameter Tuning

Hyperparameter tuning using GridSearchCV optimised XGBoost performance by adjusting learning rate, tree depth, and estimators. Initial tuning achieved strong minority-class recall, while additional class weighting slightly reduced performance because SMOTE balancing had already been applied. A refined third iteration improved stability and overall performance. However, performance differences between churn and non-churn classes remained, highlighting the difficulty of learning minority-class patterns in imbalanced datasets (He & Garcia, 2013)

3.7 Explainability Using LIME

LIME was applied to improve interpretability and explain individual churn predictions (Molnar, 2022). The analysis identified tenure, contract type, payment method, and service-related variables as major churn predictors. Longer tenure and two-year contracts reduced churn probability, supporting the reliability of the machine learning framework.

3.8 Experimental Setup

The experiment was implemented in Python using Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, and Seaborn. The dataset was split into 80% training and 20% testing sets, with feature scaling applied to Logistic Regression, KNN, and SVM. SMOTE addressed class imbalance (Chawla et al., 2002), while XGBoost class weighting showed limited improvement. Models were evaluated using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, 5-fold cross-validation, GridSearchCV, and LIME explainability.

3.9 Smote

To address class imbalance within the dataset, the Synthetic Minority Oversampling Technique (SMOTE) was applied to the training data. This process generated synthetic samples for the minority churn class to improve model learning and reduce prediction bias toward non-churn customers.

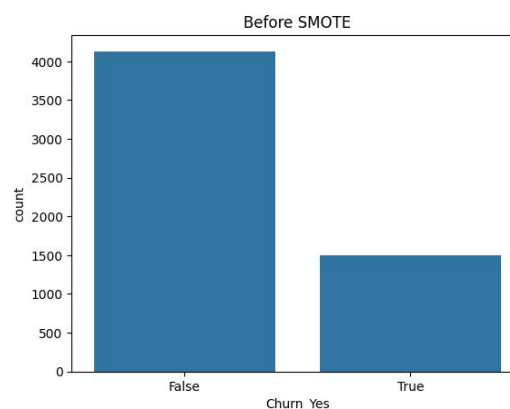


Fig 3 Distribution Before Smote

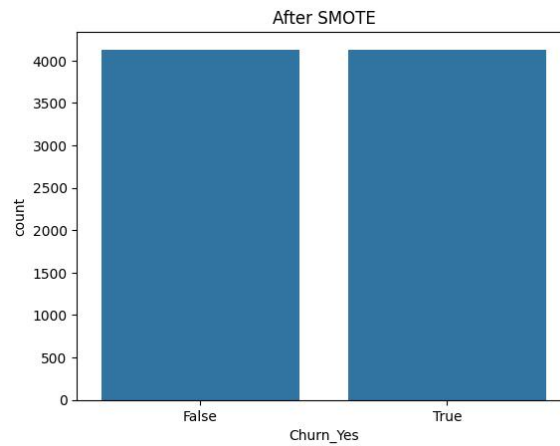


Fig 4. Distribution After Smote

To address class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was applied to the training data to generate synthetic samples for the minority churn class.

Figure 3 shows the original imbalance between churn and non-churn customers, while Figure 4 illustrates the balanced dataset after SMOTE, which improved minority class learning and churn detection.

4. Results

4.1 comparative performance of the models across the different experimental stages.

This chapter presents the evaluation and comparison of machine learning models for customer churn prediction using accuracy, precision, recall, and F1-score. Models including Logistic Regression, KNN, SVM, and XGBoost were analysed alongside SMOTE imbalance handling techniques.

Table 1 summarises the comparative performance of the models across the different experimental stages.

Model	Stage	Accuracy	Prec (0)	Prec (1)	F1 (0)	F1 (1)	Rec (0)	Rec (1)
Logistic Regression	Before SMOTE	0.79	0.83	0.62	0.86	0.56	0.89	0.52
Logistic Regression	After SMOTE	0.77	0.86	0.56	0.84	0.59	0.82	0.63
Logistic Regression	Optimized	0.77	0.86	0.56	0.84	0.60	0.82	0.63
K-Nearest Neighbours (KNN)	Initial Model	0.72	0.85	0.48	0.79	0.55	0.75	0.64
K-Nearest Neighbours (KNN)	Optimized	0.72	0.87	0.48	0.79	0.57	0.73	0.70
Support Vector Machine (SVM)	Initial Model	0.78	0.87	0.57	0.84	0.61	0.82	0.66
Support Vector Machine (SVM)	Optimized	0.76	0.85	0.54	0.83	0.57	0.81	0.61
XGBoost	Iteration 1	0.76	0.87	0.55	0.83	0.56	0.78	0.63
XGBoost	Iteration 2	0.74	0.85	0.51	0.81	0.56	0.78	0.63
XGBoost	Iteration 3	0.76	0.86	0.54	0.83	0.58	0.81	0.63

Table 1: Comparative performance of machine learning models across baseline, balanced, optimised, and hyperparameter-tuned stages.

The results showed that class imbalance significantly affected churn prediction performance. After applying SMOTE, Logistic Regression improved churn recall from 52% to 63%, while KNN optimisation increased recall but reduced precision due to more false positives.

SVM achieved one of the most balanced performances among the traditional models, and XGBoost provided the strongest balance between churn detection and overall classification stability. However, further tuning produced only minor improvements, highlighting the difficulty of fully separating churn and non-churn behavioural pattern

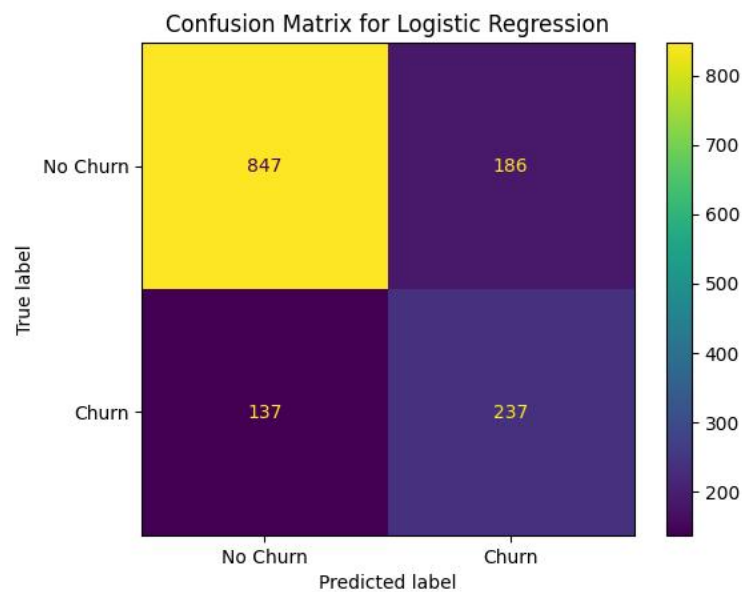


Fig 5: Confusion matrix

4.2 Hyperparameter Tuning Results

GridSearchCV hyperparameter optimisation was applied to Logistic Regression, KNN, SVM, and XGBoost to improve churn prediction performance by systematically evaluating parameter combinations. For XGBoost, multiple tuning iterations explored variations in learning rate, maximum tree depth, and number of estimators to identify optimal configurations

The selected parameters for each optimised model are presented below.

Best Parameters for Logistic Regression:
{'C': 10, 'solver': 'liblinear'}

Best Parameters for KNN:
{'metric': 'manhattan', 'n_neighbors': 7}

Best Parameters for SVM:
{'C': 10, 'kernel': 'rbf'}

Best Parameters for XGBoost Iteration 1:
{'learning_rate': 0.1, 'max_depth': 3, 'n_estimators':

200}

Best Parameters for XGBoost Iteration 2:

`{'learning_rate': 0.2, 'max_depth': 7, 'n_estimators': 200}`

Best Parameters for XGBoost Iteration 3:

`{'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 200}`

Hyperparameter optimisation was performed using GridSearchCV to improve churn prediction performance for Logistic Regression, KNN, SVM, and XGBoost models, while RandomizedSearchCV was additionally applied to Logistic Regression for comparative evaluation. The selected parameters for each optimised model are presented below.

4.3 Cross Validation

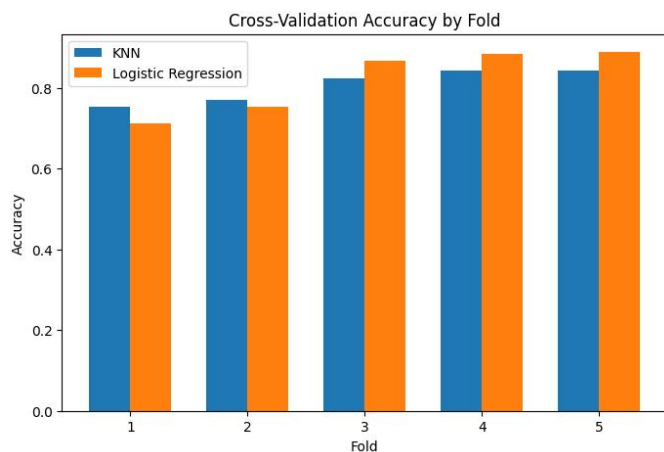


Fig.6

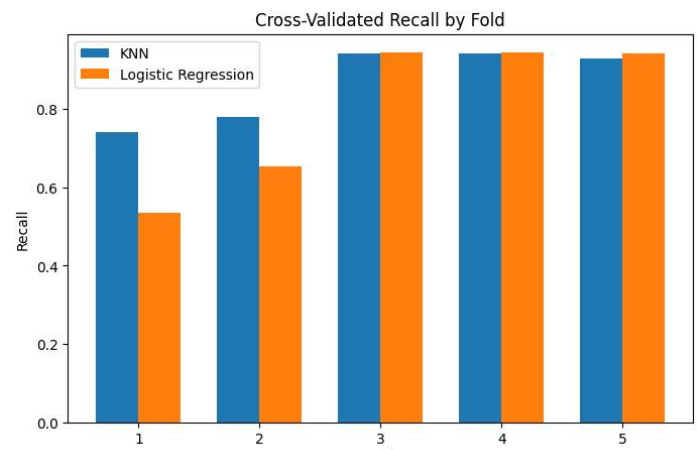


Fig.7

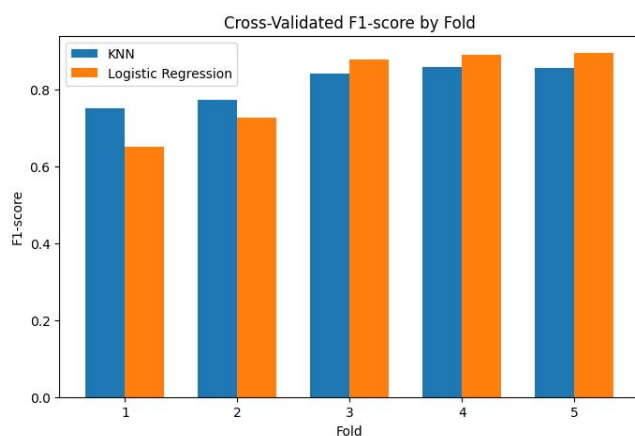


Fig 8

To evaluate model stability and generalisation, 5-fold cross-validation was conducted using Logistic Regression and KNN with accuracy, recall, and F1-score metrics.

The results showed that Logistic Regression achieved more consistent and stable performance across the folds, particularly in recall and F1-score, while KNN demonstrated greater variation and sensitivity to training data differences. Overall, Logistic Regression provided better generalisation capability for customer churn prediction on the imbalanced dataset.

4.4 ROC Curve Analysis

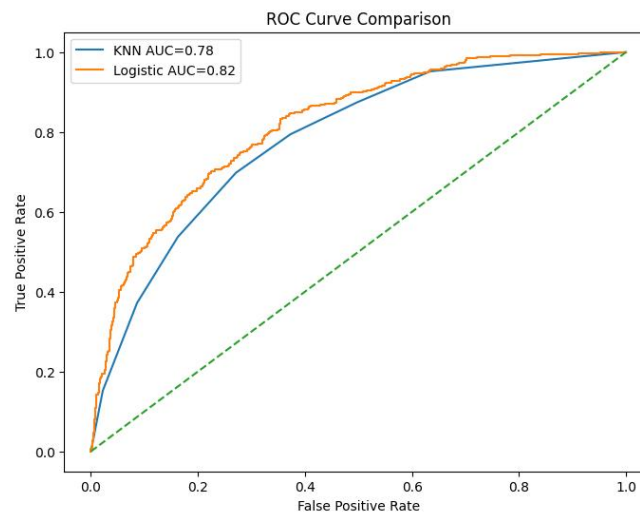


Fig.9 ROC CURVE

ROC analysis evaluated Logistic Regression and KNN using AUC scores across classification thresholds. Logistic Regression achieved a higher AUC score of 0.82 than KNN's 0.78, demonstrating stronger sensitivity, better churn discrimination, and more reliable overall classification performance for customer churn prediction.

4.4 LIME Explainability Analysis

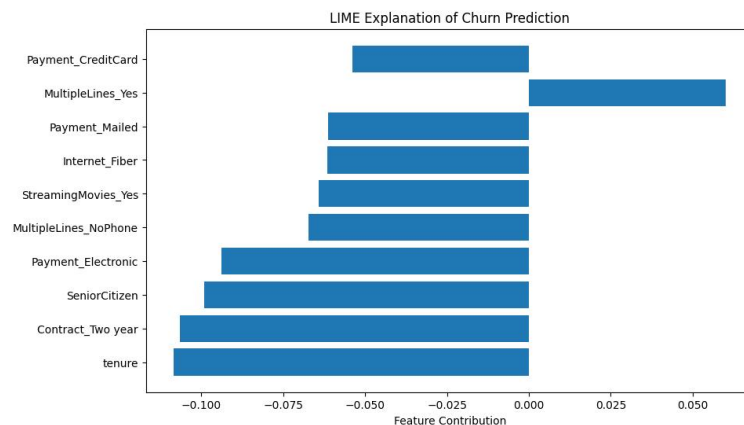


Fig 10. Lime Graph

LIME was applied to explain individual churn predictions. The results showed that tenure and Contract_Two year reduced churn likelihood, while MultipleLines_Yes increased churn prediction. Features such as Internet_Fiber and StreamingMovies_Yes had smaller influences, improving model transparency and interpretability.

4.5 Experimental Improvement Strategy

Initial baseline experiments showed that the dataset suffered from class imbalance, where non-churn customers significantly outnumbered churn customers. This affected the model's ability to correctly identify minority churn cases, particularly in recall performance. To address this issue, SMOTE was applied to balance the training data and improve churn detection.

Following balancing, hyperparameter optimisation was performed using GridSearchCV to improve model performance through systematic parameter tuning. Cross-validation was then conducted to evaluate model stability and generalisation across multiple data folds rather than relying on a single train-test split.

Finally, explainability techniques including feature importance analysis and LIME were implemented to improve model interpretability by identifying the features most responsible for churn predictions. These steps enhanced the reliability, transparency, and overall predictive performance of the developed machine learning pipeline.

5. Discussion and Conclusion

This study demonstrated that machine learning models can effectively predict customer churn within the telecommunications sector. Logistic Regression, KNN, SVM, and XGBoost were evaluated using accuracy, precision, recall, F1-score, ROC analysis, and cross-validation. Among the models, Logistic Regression achieved the most stable and balanced performance, while SMOTE improved churn recall from 0.52 to 0.63, highlighting the trade-off between accuracy and minority-class detection. Feature analysis identified tenure, contract type, internet service, payment method, and technical support as key factors influencing churn behaviour. Cross-validation, ROC analysis, and LIME explainability further confirmed the reliability and interpretability of the developed models, with Logistic Regression achieving the highest stability and an AUC score of 0.82.

Despite SMOTE, hyperparameter tuning, and XGBoost optimisation, perfectly balancing churn and non-churn prediction remained difficult due to similarities between churn and retained customer behaviours. Although churn recall improved, models still favoured the majority class, reinforcing the importance of prioritising churn detection over overall accuracy..

From an ethical and professional perspective, customer churn prediction systems must ensure fairness, transparency, and compliance with data protection regulations such as GDPR. Explainable AI techniques are important for improving trust and accountability in automated decision-making systems.

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